

1. List the year and title of each book

$\Pi_{year, Title}(BOOKS)$

```
SELECT B.year, B.title
FROM Books B
```

2. List all information about students whose major is CS

$\sigma_{Major='CS'}(STUDENT)$

```
SELECT S.StId, S.StName, S.Major, S.Age
FROM Students S
WHERE S.Major = 'CS'
```

3. List all students with books they can borrow

$Students \bowtie Borrows \bowtie Books$

```
SELECT *
FROM Students S, Borrows B1, Books B2
WHERE S.StID = B1.StId AND B1.DocId = B2.DocId
```

4. List all books published by McGraw-Hill before 1990

$\sigma_{Publisher='McGraw-Hill' \text{ AND } Year < 1990}(BOOKS)$

```
SELECT *
FROM Books B
WHERE B.Publisher = 'McGraw-Hill' AND B.Year < 1990
```

5. List the name of those authors who are living in Davis

$\Pi_{AName} \sigma_{Address='Davis'}(AUTHORS)$

```
SELECT A.AName
FROM Authors A
WHERE Address = 'Davis'
```

6. List the name of students who are older than 30 and who are not studying CS

$\Pi_{StName}(\sigma_{Age > 30 \text{ AND } Major \neq 'CS'}(STUDENTS))$

```
SELECT S.StName
FROM Students S
WHERE S.Age > 30 AND S.Major != 'CS'
```

7. Rename AName in the relation AUTHORS to Name

$\rho_{AUTHORS(NAME, ADDRESS)}(AUTHORS)$

SELECT AName AS Name
FROM Authors

8. List the names of all students who have borrowed a book and who are CS major

$\Pi_{StName}(\sigma_{Major = "CS"}(STUDENTS \bowtie BORROWS))$

SELECT S.StName
FROM Students S, Borrows B
WHERE S.Major = 'CS' AND S.StId = B.StId

SELECT S.StName
FROM Students S
WHERE Major = 'CS'

9. List the title of books written by the author “Jones”

$\Pi_{Title}(\sigma_{AName = 'Jones'}(BOOKS \bowtie haswritten))$

$\Pi_{Title}(\sigma_{AName = 'Jones'}(BOOKS \bowtie haswritten))$

SELECT B.Title
FROM Books B
WHERE B.DocId IN (SELECT DocID
FROM Has_Written H
WHERE AName = 'Jones')

10. As previous, but not books that have the keyword “database”

$\Pi_{Title}((\sigma_{AName = 'Jones'}(Books \bowtie HasWritten) - (\sigma_{Keyword = 'database'}(Book \bowtie describes))))$

SELECT Title
FROM Books
WHERE DocID IN (SELECT DocID
FROM has-written
WHERE AName = 'Jones')
AND DocID NOT IN (SELECT DocID
FROM describes
WHERE keyword = 'database')

11. Find the name of the youngest student

$\Pi_{StName}(\sigma_{Age=\min(Age)}(STUDENTS))$

SELECT StName

FROM Students

WHERE Age = (SELECT Min(Age)

FROM Students)

12. Find the title of the oldest book

$\Pi_{Title}(\sigma_{Year=\min(Year)}(Books))$

SELECT Title

FROM Books

WHERE Year = (SELECT Min(Year)

FROM Books)